

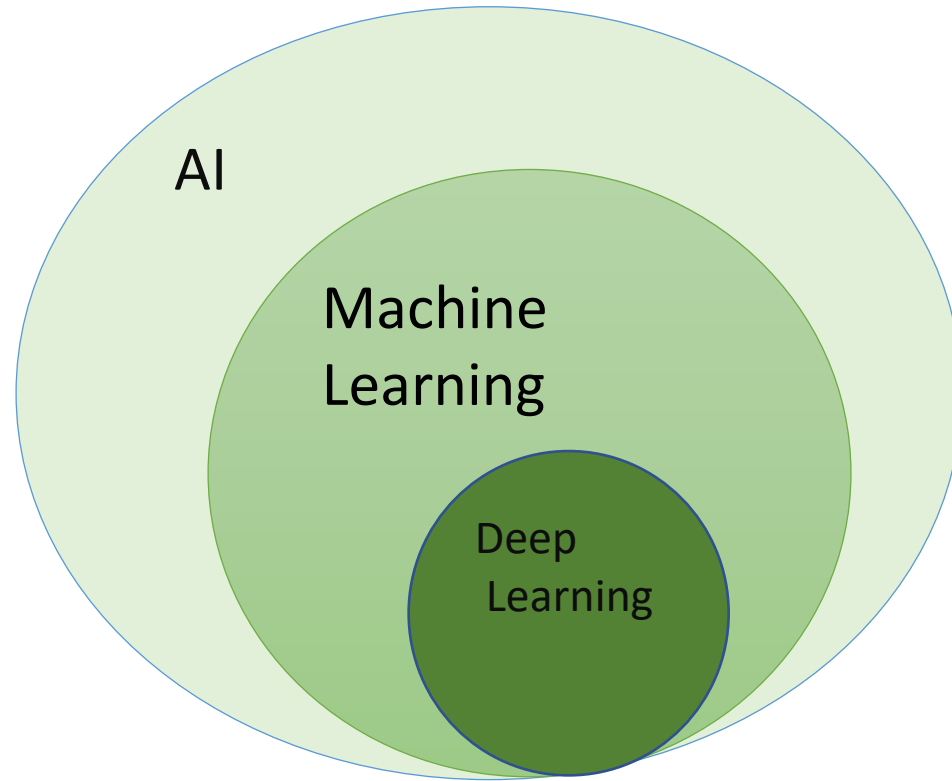


Introduction to Machine Learning (CS-436/536)

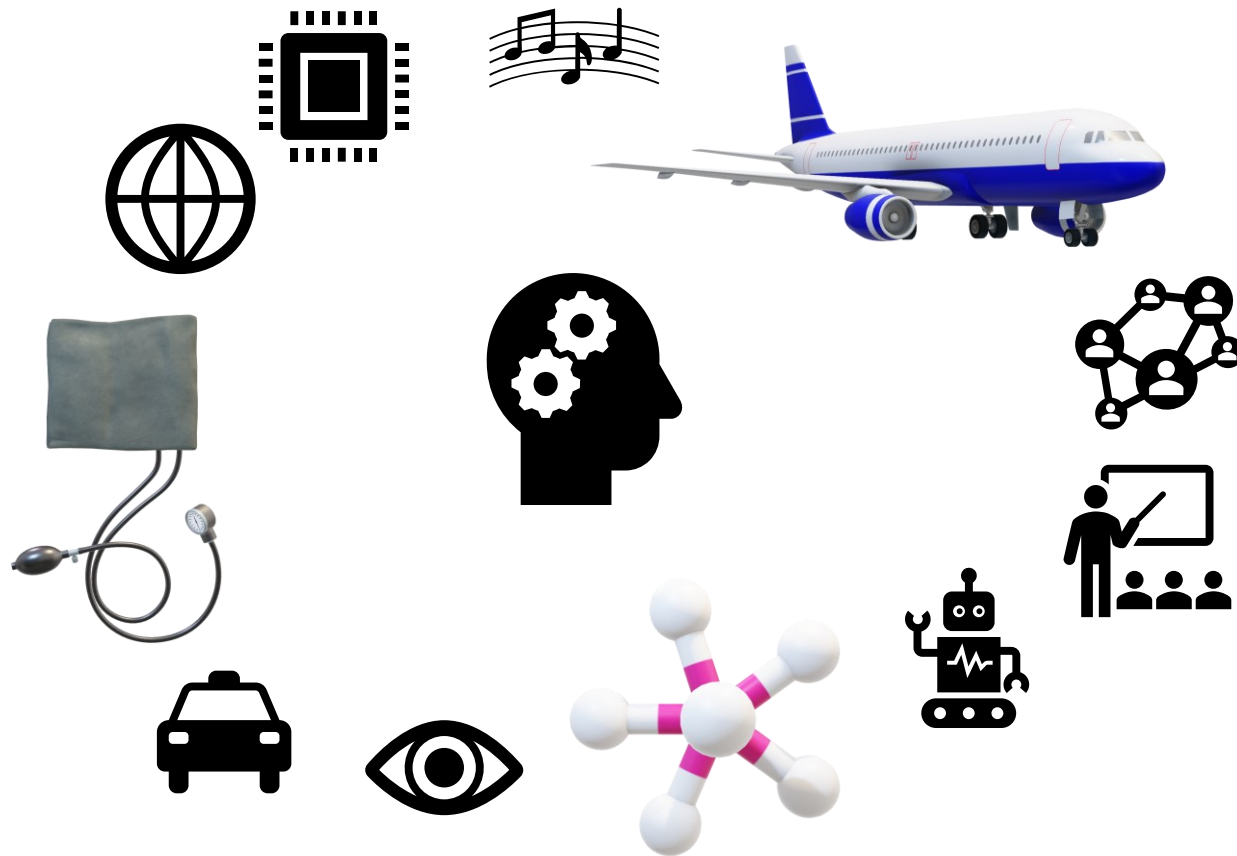
Lecture-1: Introduction

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Machine Learning



AI is the new electricity – Andrew Ng



Pre-requisite:

- Mathematics (Linear Algebra, Calculus, Statistics)
- Programming Language (Python)
- Deep Learning Framework (**PyTorch**/TensorFlow)

* Don't worry! we will go over some mathematics and programming basics in the next lecture.

Grading Policy:

- Everyday Quiz (5%)
- Take-Home Assignments (20%)
- Paper Critique (10%)
- Final Project (20%)
- Class Test/Exam (30%)
- Final Exam (15%)

Everyday Quizzes (5 %):

- Everyday after/before the class lecture
- 14/15 quizzes
- Forget about grade focus on the questions for the quiz
- Sometimes they will give enough hints for the major quizzes
- Attendance Required (85 % ~ 12/15)

Take-Home Assignments (20 %):

- Total Five Assignments
- Each assignment weight is 5 % of the total grades
- Best 4 counts towards the final grade
- Mainly coding problems in Python/PyTorch

Paper Critique (10 %):

- I will give you papers and a list of task to complete.
- Make sure you read and understand the paper.
- Present your critique/summary in a logical flow (format will be given).

Journals and Conferences to Follow (on or After 2019):

- ❑ CVPR · IEEE/CVF Conference on Computer Vision and Pattern Recognition
- ❑ ECCV · European Conference on Computer Vision
- ❑ ICLR · International Conference on Learning Representations
- ❑ NeurIPS · Neural Information Processing Systems
- ❑ ICML · International Conference on Machine Learning
- ❑ AAAI · AAAI Conference on Artificial Intelligence
- ❑ CHI · ACM Conference on Human Factors in Computing Systems
- ❑ SIGKDD · ACM International Conference on Knowledge Discovery and Data Mining
- ❑ ICCV · International Conference on Computer Vision
- ❑ ASPLOS, MICRO, ISCA, USENIX Security, IEEE S&P, IJCAI, IJCNN, DAC, EMNL, ACL, NAACL

Class Tests (30 %):

- Exam will be on all the topics covered before the exam classes
- Only short questions and conceptual test
- Exam will include mathematical problems
- No programming questions
- Best four out of five
- Each exam weight is 7.5 %

Final Project (20 %)

- Three Major Types of Project:

- 1) Reproduce an existing paper and then perform additional analysis.**

- 2) Apply an ML algorithm on a new dataset.

- 3) Reproduce an existing paper and improve its methodology.

- First two are very typical and will receive regular grades.
- The last one is encouraged for bonus points.

Final Exam (15 %):

- Exam will include only the topics covered before the exam classes from the last mid-term quiz.
- Only short questions and conceptual test
- Exam will include mathematical problems
- No programming questions

Final Project (20 %):

- Group of two
- Pick any topic of machine learning
- Pick a research paper in selected conferences
- Read it and make sure it has **open source code**
- Make sure you understand the paper and their code.
- Present your project (Final Project Demo) (10 %= 8+2%).
- Final Project report due before the final exam week (10 %).
- Example:

Weekly Lectures:

Week	Date	Lecture?	Topics
1	8/18/2025 (Tuesday)	Yes	Introduction (Lecture-1)
	8/20/2025 (Thursday)	Yes	Machine Learning Basics (Lecture-2)
2	8/25/2025 (Tuesday)	Yes	Mathematical Foundation (Lecture-3)
	8/27/2025 (Thursday)	Yes	Mathematical Foundation (Lecture-3) Code Basics (Python and PyTorch) (Lecture-4)
3	9/1/2025 (Tuesday)	Yes	Quiz-1
	9/4/2025 (Thursday)	Yes	Linear Regression (Lecture-5)

Weekly Lectures:

Week	Date	Lecture?	Topics
4	9/8/2025 (Tuesday)	No	Monday Class meet
	9/10/2025 (Thursday)	Yes	Linear Regression (Lecture-6)
5	9/15/2025 (Tuesday)	Yes	Logistic Regression (Lecture-7)
	9/17/2025 (Thursday)	Yes	Evaluation (Lecture-8)
6	9/23/2025 (Tuesday)	yes	Quiz-2
	9/25/2025 (Thursday)	Yes	Neural Network and Backpropagation (Lecture 9 & 10)

Weekly Lectures :

Week	Date	Lecture?	Topics
7	9/29/2025 (Tuesday)	Yes	Neural Network and Backpropagation (Lecture 9 & 10) Deep Neural Network (Lecture 11 & 12)
	10/1/2025 (Thursday)	Yes	Deep Neural Network (Lecture 11 & 12)
8	10/7/2025 (Tuesday)	Yes	Quiz-3
	10/9/2025 (Thursday)	Yes	Support Vector Machine (Lecture-13)
9	10/13/2025 (Tuesday)	No	Fall Break
	10/15/2025 (Thursday)	No	Fall Break

Weekly Lectures :

Week	Date	Lecture?	Topics
10	10/20/2025 (Tuesday)	Yes	Probability Based Learning (Lecture-14)
	10/22/2025 (Thursday)	Yes	Decision Trees and Clustering (Lecture 15 & 16)
11	10/27/2025 (Tuesday)	Yes	GAN and Self-supervised learning (Lecture 15 & 16) Dimensionality Reduction (Lecture-17)
	10/29/2025 (Thursday)	Yes	Quiz-4
12	11/3/2025 (Tuesday)	Yes	Dimensionality Reduction (Lecture-17) Hidden Markov Model (Lecture-18)
	11/5/2025 (Thursday)	Yes	Advanced topics: LLMs and AI Security (Lecture-19)

Weekly Lectures :

Week	Date	Lecture?	Topics
13	11/10/2025 (Tuesday)	Yes	Final Project Demo
	11/12/2025 (Thursday)	Yes	Final Project Demo
14	11/17/2025 (Tuesday)	Yes	Final Project Demo
	11/19/2025 (Thursday)	Yes	Quiz-5
15	11/24/2025 (Tuesday)	No	Friday's classes meet
	11/26/2025 (Thursday)	No	Thanksgiving break
16	12/1/2025 (Tuesday)	Yes	Final Project Demo
	12/3/2025 (Tuesday)	Yes	Final Project Demo

Class Rules

- Deadlines are strict
- Late Assignment submission 20% deduction for each day.
- You are allowed to have 2 grace day for the entire semester for assignments mostly.
- Copying or plagiarism will cost the grade of the entire assignment to be zero
- Second incident will lower your course grade by 2-levels
- Third incident will result in 'F' and will be reported to the dean's office
- Attend the classes regularly (key to your grade)
- Follow the university and department guidelines for COVID and others
- Send me emails with any inquiry (include CS 436/536 in subject line)
- Class lectures will be available online after the class
- **Grace day can not be used for the Final Project or Exam**

Objectives of The Course:

- Build solid foundation of Machine Learning.
- Apply ML to solve real world problems.
- Gain familiarity with advanced ML algorithms in the literature.
- Learn to read SOTA ML papers.
- Build a project on Machine Learning.
- Learn to use advanced Machine Learning tools (e.g., PyTorch).

Office Hours and TA:

- Office: Engineering Building (EB) N-30
- Office Hours: T/TR 10:00am-11:00am
- Default: In-Person
- TA: will be announced this week

Some Textbook References:

- A. *FUNDAMENTALS OF MACHINE LEARNING FOR PREDICTIVE DATA ANALYTICS. Algorithms, Worked Examples, and Case Studies* John by D. Kelleher Brian Mac Namee Aoife D'Arcy (Permission taken MIT Press)
- B. *Dive into Deep Learning*, Aston Zhang, Zachary C. Lipton, Mu Li, and Alexander J. Smola (July 25, 2021)
- C. *Master Machine Learning Algorithms* (Jason Brown Lee)
- D. *Machine Learning* by Tom Mitchell (Dr. Mitchell has shared a free copy of his book with me for you all to use for the study purposes: [Machine - Learning - Tom Mitchell.pdf \(cmu.edu\)](#)). Thanks to Dr. Mitchell.
- E. *Pattern Recognition and Machine Learning* by Christopher M. Bishop

Journals and Conferences to Follow:

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- ☐ SIGKDD · ACM International Conference on Knowledge Discovery and Data Mining
- ☐ ICCV · International Conference on Computer Vision
- ☐ Arxiv papers (Not published)

Other sources:

- ❑ Open Source dataset: MNIST, Fashion-MNIST, CIFAR-10 & CIFAR-100
- ❑ Kaggle (dataset source and competition)
- ❑ GitHub (Code)
- ❑ Google Colab (Simulation Platform)

Simulation Platform

- Google Colab (Free but limited)
- Use Colab Pro which you can sign up as student get free access.
- Your own Laptop/GPUs
- For Final Project: T4 GPU nodes available (2 nodes)

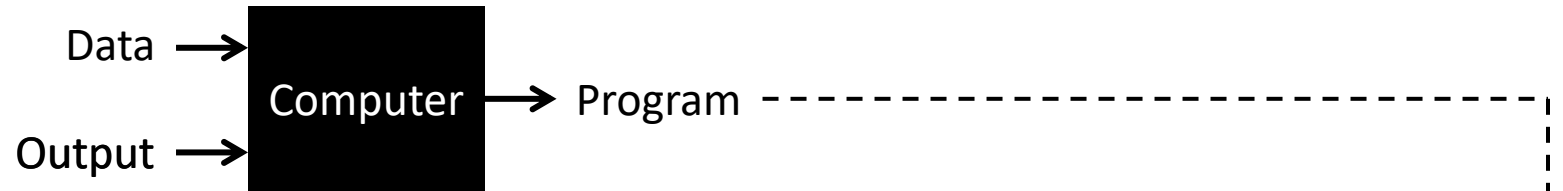
openhpc.binghamton.edu

- Use your Binghamton ID and password to login.
- Instructions: [Welcome to Spiedie Docs! | Spiedie | Binghamton University](#)
(same for OPEN HPC)
- Contact: Phil (pvalenta@binghamton.edu)
- Submit a ticket: cs-ticket@binghamton.edu

What is Machine Learning?

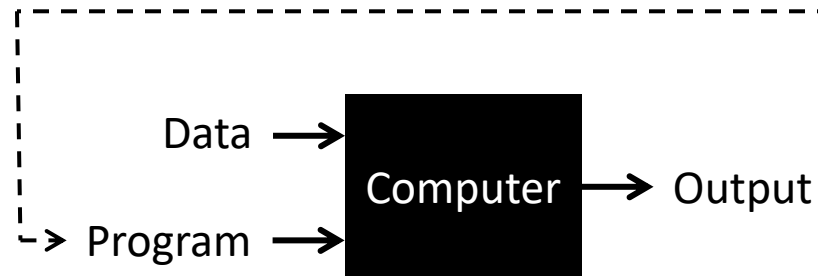
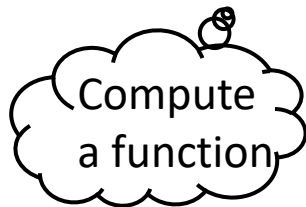
“computer programs that automatically improve their performance through experience” – Herbert Simon

Machine Learning:



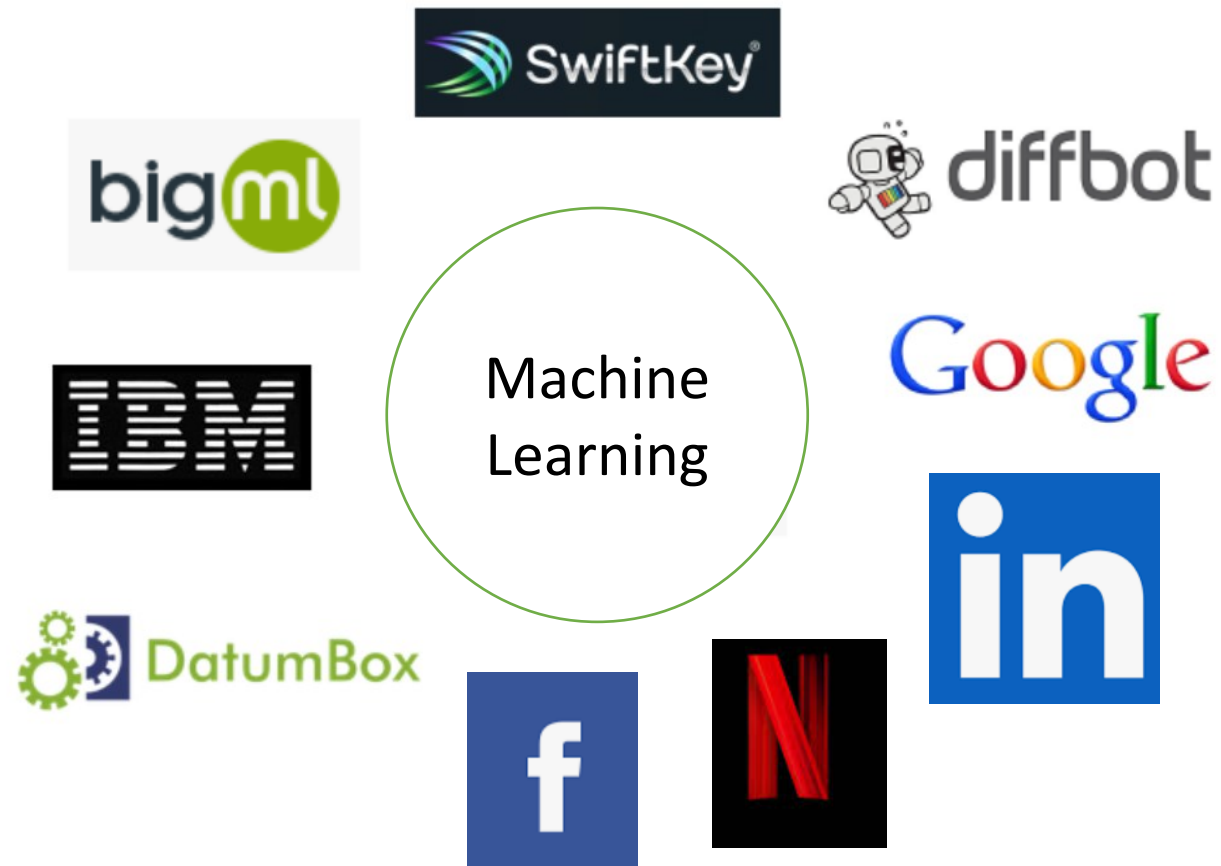
“... the ability to learn without being explicitly programmed” – Arthur Samuel

Traditional Programming:

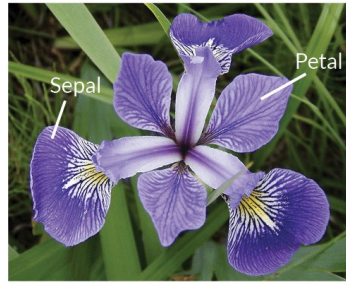
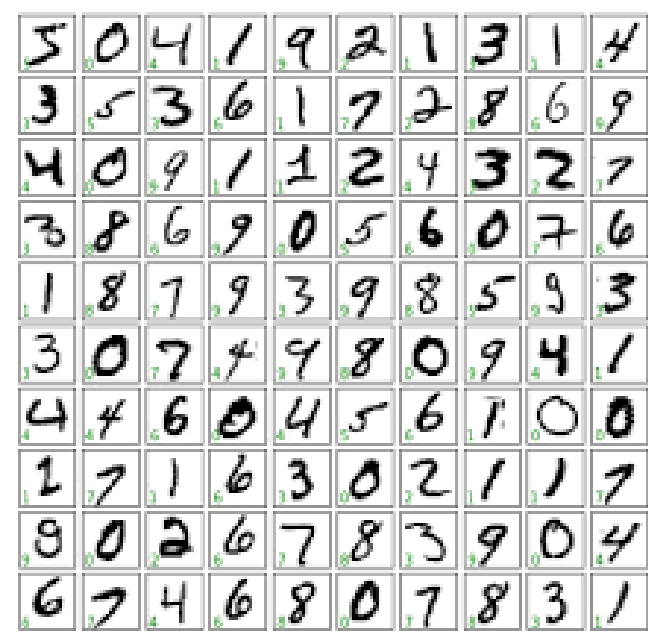


What it is not:
The design approach

Machine learning in Real Life



Learning to Solve Hard Problems with Data



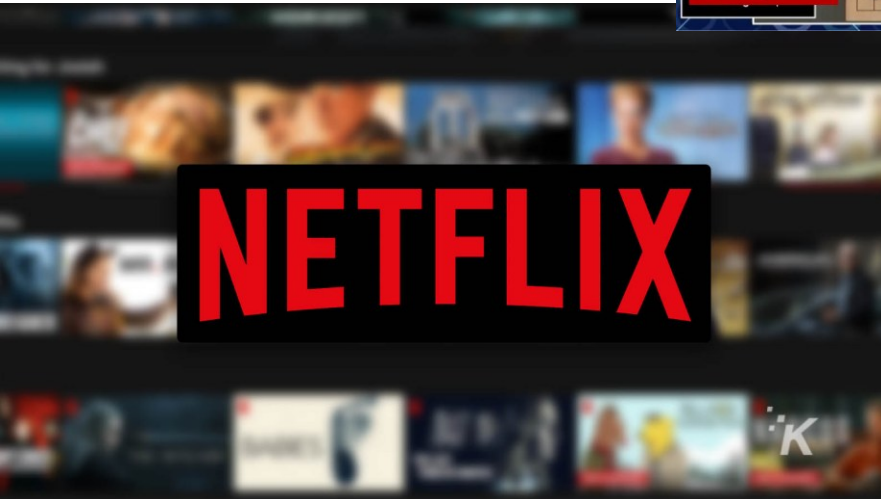
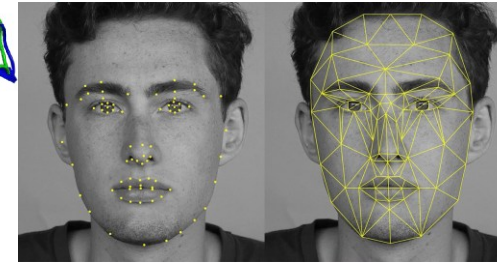
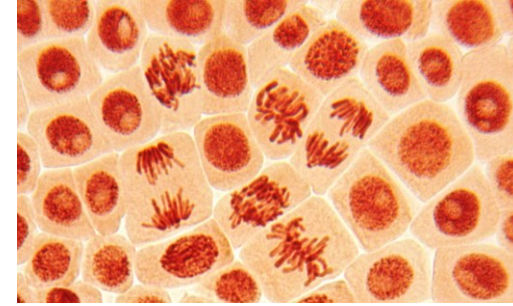
Iris Versicolor



Iris Setosa



Iris Virginica



- 🔍 how to
- 🔍 how to **care for a ficus tree**
- 🔍 how to **propagate ficus elastica**
- 🔍 how to **propagate ficus benjamina**
- 🔍 how to **get job in machine learning**

When is Machine Learning Useful?

- Human expertise does not exist (navigating on Mars, protein structure)
- Humans can't explain their expertise (speech / face recognition)
- Models must be customized (personalized medicine)
- Models are based on huge amounts of data (genomics)
- No “analytical” solution

Why ML ?

- Data is Abundant
- Knowledge is difficult
- Can we generalize the data ?
- Advantage: Use prior knowledge to predict the future
- Machine learning model
- Example: Face Detection (Iphone)
- Example: Spam detector (GMAIL)

Types of Machine Learning

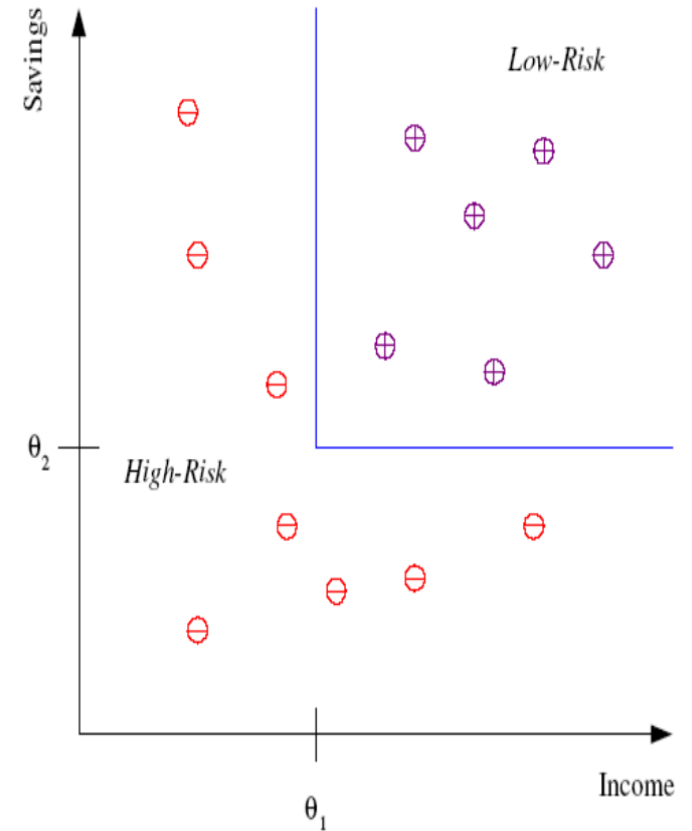
- ☐ Supervised Learning
- ☐ Unsupervised Learning
- ☐ Semi-supervised Learning
- ☐ Self-supervised Learning
- ☐ Reinforcement Learning

Supervised Learning

Classification:

Example: Credit Score

Low Risk: Income higher than θ_1 and Savings higher than θ_2

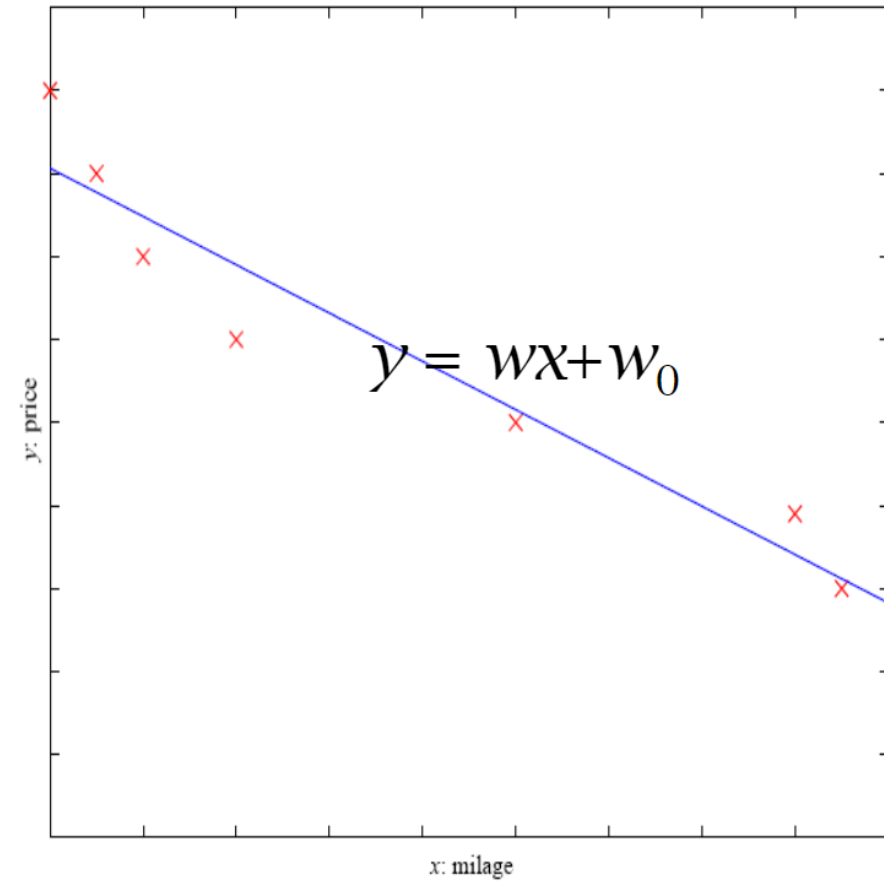


Supervised Learning:

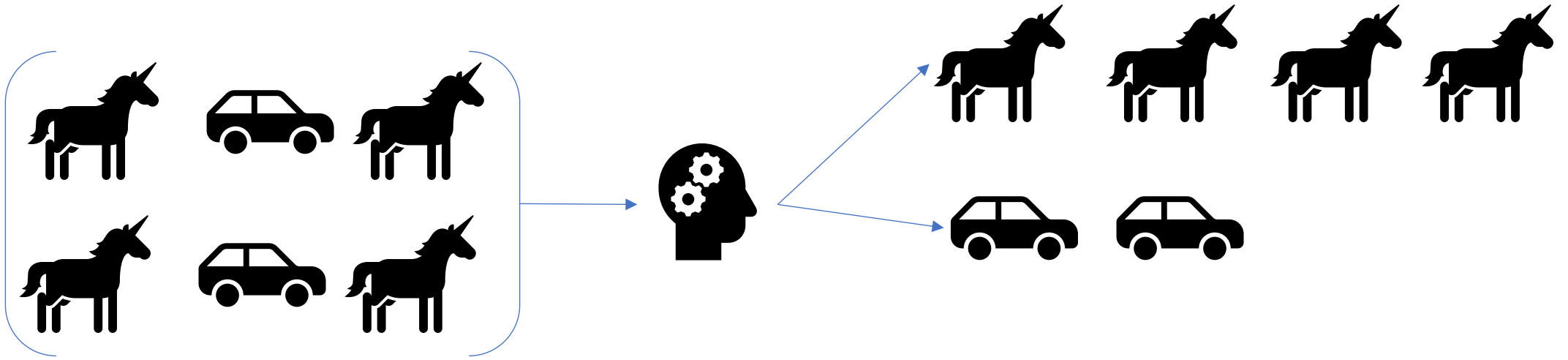
Regression:

Example: Used Car Price

Price of car (y) Changes with milage (x)



Un-supervised Learning:



Examples: Clustering, Association and so on.

Other Learning Types:

- ❑ Semi-supervised (Partially guided e.g., GAN)
- ❑ Self-supervised (Generates it's own label e.g., contrastive learning)
- ❑ Reinforcement Learning (Reward based learning e.g., Q-Learning)

Some Applications of Machine Learning:

- ❑ Speech recognition,
- ❑ Natural language processing
- ❑ **Computer vision**
- ❑ Medical outcomes analysis
- ❑ Robot control
- ❑ Computational biology

Finance: Stock Prices

Market Summary > S&P 500

3,960.97

+3,814.32 (2,600.97%) ↑ all time

Jan 13, 9:50 AM EST • Disclaimer

+ Follow

1D 5D 1M 6M YTD 1Y 5Y Max

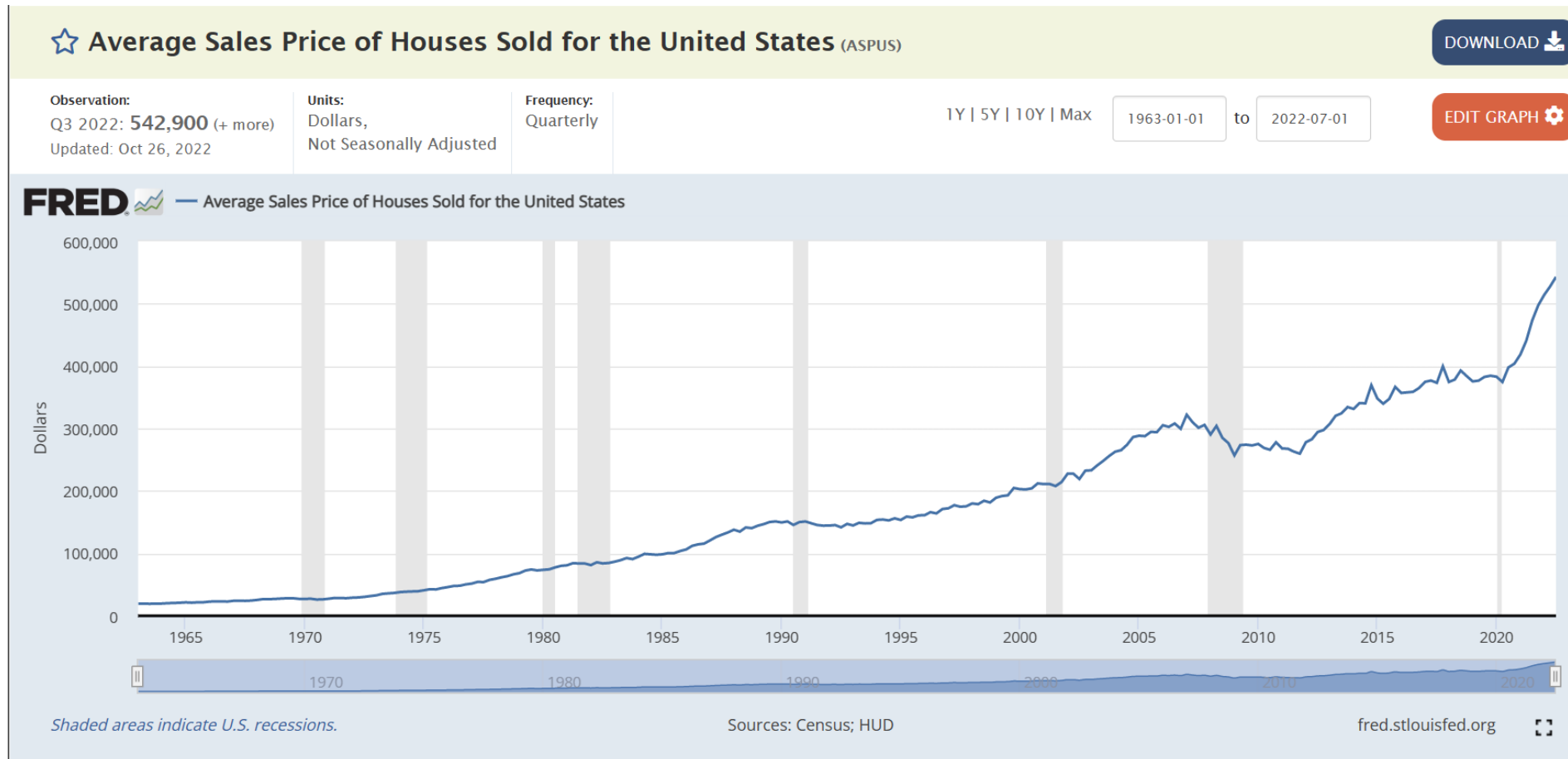


Open	3,960.60	Low	3,947.67	52-wk high	4,744.13
High	3,961.40	Prev close	3,983.17	52-wk low	3,491.58

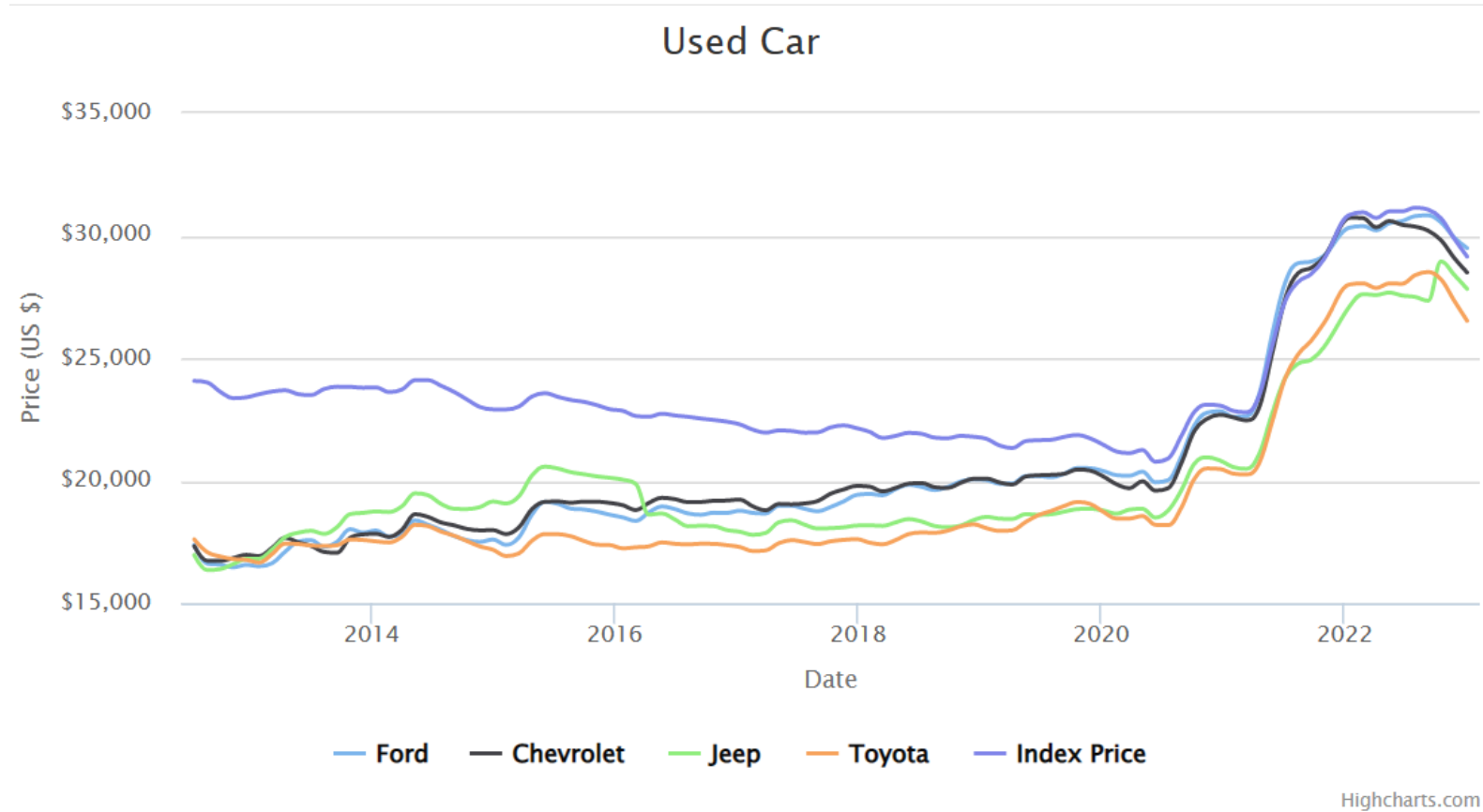
More about S&P 500 →

Feedback

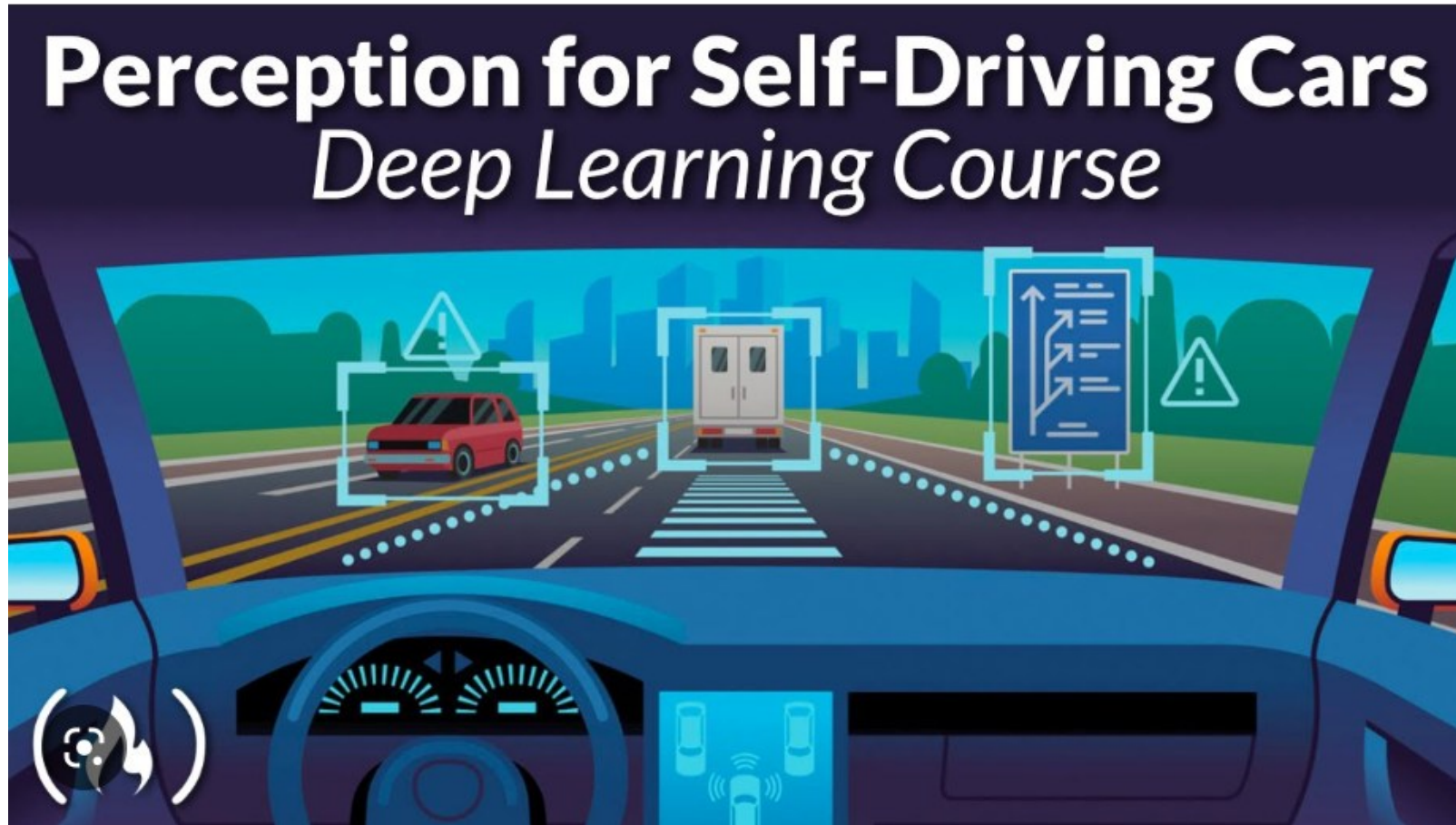
Finance: House Prices



Finance: Car Prices



Transportation: Self-Driving Cars



Language Processing: Chat GPT

- Translation
- Summarization
- Answer Random Questions
- Text to Speech and Speech to text
- Write Novels
- Content Creation

What Else?

- Home Work
- Assignments
- Take Home Exams

Quiz:

- Why did you take the course?
- What is your learning expectation from the course?
- Use your imagination to write down few problems that we may solve using machine learning in future?

Acknowledgements

- Thanks to Dr. Baoxin Li at Arizona State University, Dr. Aston Zhang at Amazon and Dr. Sujoy Sikdar at Binghamton University for giving permission to use their books and slide contents. Thanks to Dr. Mitchell at CMU for sharing his book for free. Special thanks to MIT press for granting educational permission to use contents of their book (FUNDAMENTALS OF MACHINE LEARNING FOR PREDICTIVE DATA ANALYTICS)